

Technologies and Tools Used

This document contains a list of the technologies, libraries, and development tools used during different phases of the project, including analysis, design, implementation, and testing, together with their versions and purpose.

1. Programming Languages

Python 3.11.0

- Used in an earlier Django development environment.
- Purpose: implementation of the backend part of the application during an earlier project phase.

Python 3.12.10

- Used in the final project version.
- Purpose: implementation of the backend system.

Java 17

- Used for automated GUI testing with Selenium and TestNG.

SQL (MySQL dialect)

- Used through database scripts and configuration.
- Purpose: creation, insertion, and management of relational database data.

HTML5

- Used in Django templates and UI prototypes.
- Purpose: structure of the user interface.

CSS3

- Used for styling the user interface and prototypes.

2. Backend Technologies and Frameworks

Django 5.2.x

- Main backend framework used in the project.
- Used for request routing, user authentication, ORM and database interaction, database migrations, and template rendering.
- The project was primarily developed using Django 5.2.x.

3. Database

MySQL 8.0.41

- Configured as the main relational database.
- Used for storing system data such as users, patients, appointments, interventions, and related information.

mysqlclient 2.2.7

- Python package used to connect the Django application to MySQL.

4. Python Libraries

asgiref 3.11.0

- Provides ASGI support and asynchronous functionality for Django.

sqlparse 0.5.3 / 0.5.4

- Used internally by Django for parsing SQL statements.

tzdata 2025.2

- Used for time zone management.

Pillow 12.0.0

- Used for image processing, uploads, and image display.

pip 22.3 / setuptools 65.5.0

- Used for Python package and dependency management.

5. Frontend Technologies

Django Template Engine

- Used to generate dynamic server-rendered HTML content.

JavaScript (ES6+)

- Used for dynamic user interface behavior such as menus, filtering, carousels, and other interactive elements.

AJAX / Fetch API

- Used for asynchronous communication with the backend without reloading the page.
- Used for operations such as retrieving available appointments and making reservations.

HTML/CSS Prototypes

- Used during the interface design and prototyping phase.

6. Testing

Selenium 4.35.0

- Used for automated web application testing.

TestNG 6.8.1

- Used for organizing and executing automated test cases.

Failsafe 3.3.2

- Used to support test execution during the build process.

Apache Maven

- Used for Java dependency management, build processes, and running automated tests.

GeckoDriver 0.36.0

- Used to control Firefox during Selenium tests.

Django Test Framework 5.2.x

- Used for unit testing Django models, views, and business logic.

7. Development Tools and Environments

Python Virtual Environment (venv)

- Used to isolate project dependencies.

JetBrains IDEs

- PyCharm / IntelliJ IDEA.
- Used for project development and management.

Microsoft Word

- Used for database specification and documentation.

PDF Documents

- Used for documenting requirements, scenarios, and system functionality.

Diagrams (PNG/JPG)

- Used for ER diagrams and other system/database models.